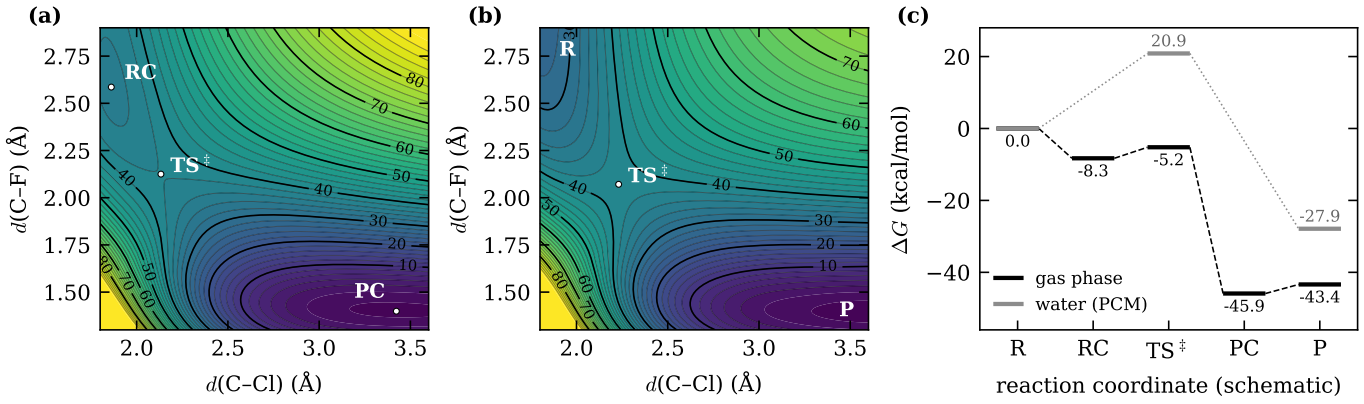

(a)
RC
TS‡
PC
d(C–F) (Å)
d(C–Cl) (Å)
(b)
R
TS‡
P
d(C–F) (Å)
d(C–Cl) (Å)
(c)
20.9
0.0
-8.3
-5.2
-27.9
-45.9
-43.4
gas phase
water (PCM)
ΔG (kcal/mol)
R
RC
TS‡
PC
P
reaction coordinate (schematic)